The MOND Acceleration Scale from Rindler–de Sitter Thermodynamics

A geometric derivation of $a_0 = cH/2\pi$

B. ten Broek

Independent researcher, Netherlands
ORCID: 0009-0002-6546-0588

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Abstract

The MOND acceleration scale a_0 is empirically required to explain galactic dynamics but has historically lacked a theoretical foundation. We show that a_0 follows from the thermodynamics of de Sitter space: if the Rindler horizon of a uniformly accelerated observer equals the Gibbons–Hawking thermal wavelength, then $a_0 = cH/2\pi$. The factor 2π enters as the KMS period of the de Sitter Euclidean propagator. This identification is a physically motivated postulate, not a mathematical necessity.

The interpolating function $\mu(x) = x/\sqrt{1+x^2}$ is then derived from the Deser–Levin temperature addition law in de Sitter space, and the covariant action follows uniquely: $\mathcal{F}(Q) = \sqrt{Q+Q^2} - \operatorname{arcsinh}(\sqrt{Q})$, satisfying ghost-free stability conditions for all $Q > 0$. The stress-energy tensor satisfies the NEC and WEC, reduces to the Maxwell gravitational stress in the Newtonian limit, and yields the exact BTFR $V^4 = GM_b a_0$ in the deep-MOND limit. With a_0 as the single calibration parameter, the derived $\mu(x)$ fits the SPARC radial acceleration relation McGaugh, Lelli & Schombert (2016); the theoretical prediction $cH_{dS}/2\pi = 1.13 \times 10^{-10}$ m s^{-2} falls 6% below the SPARC best-fit 1.20×10^{-10} m s^{-2}, consistent with the Hubble tension. Gravitational lensing requires a disformal metric completion and is deferred to future work.

Introduction

The MOND empirical law of Milgrom Milgrom (1983) introduces a critical acceleration $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} below which galactic dynamics depart from Newtonian predictions. The Radial Acceleration Relation (RAR) McGaugh, Lelli & Schombert (2016) confirms this across hundreds of galaxies spanning five orders of magnitude in baryonic mass.

Milgrom noted $a_0 \sim cH_0/2\pi$ but offered no derivation of the 2π. McCulloch McCulloch (2007) derived MOND-like behaviour from Unruh radiation truncated by the Hubble horizon, but without a covariant field theory or a derivation of the interpolating function. Milgrom Milgrom (2020) discussed the a_0–Λ connection at the phenomenological level and noted $a_0 \simeq cH_{dS}/2\pi$, but did not derive the 2π factor, the functional form of $\mu(x)$, or a covariant action. The present paper provides all three, and additionally shows that constant a_0 (H2) is a structural consequence of the action, not an independent assumption:

1. **Geometric identification** (Section 2): $a_0 = cH/2\pi$ from the condition $d_R = \lambda_{T_H}$, with the 2π from the KMS period of the de Sitter Euclidean propagator.
2. **Interpolating function** (Section 3): $\mu(x) = x/\sqrt{1+x^2}$ from the Deser–Levin temperature addition law.
3. **Covariant action** (Section 4): unique ghost-free $\mathcal{F}(Q) = \sqrt{Q+Q^2} - \operatorname{arcsinh}(\sqrt{Q})$.

The connection between detector thermodynamics and gravitational field equations is postulated, not derived from first principles.

The Rindler–de Sitter Threshold

Gibbons–Hawking Temperature and Thermal Wavelength

De Sitter spacetime with Hubble rate H carries the Gibbons–Hawking (GH) temperature Gibbons & Hawking (1977):

$$T_H = \frac{\hbar H}{2\pi k_B}. \tag{1}$$

The factor 2π is the circumference of the Euclidean time circle that enforces regularity at the cosmological horizon (KMS condition). The thermal de Broglie wavelength of this vacuum is:

$$\lambda_{T_H} = \frac{\hbar c}{k_B T_H} = \frac{2\pi c}{H}. \tag{2}$$

The Rindler Horizon

An observer with proper acceleration a has a Rindler horizon at:

$$d_R = \frac{c^2}{a}. \tag{3}$$

The Threshold

Theorem 1 (Rindler–de Sitter Threshold)**.** *If the physically relevant decoherence condition is $d_R = \lambda_{T_H}$ — the Rindler horizon of a freely-falling observer equals the thermal de Broglie wavelength of the de Sitter vacuum — then*

$$\boxed{a_0 = \frac{cH}{2\pi}.} \tag{4}$$

The factor 2π enters as the KMS period of the de Sitter Euclidean propagator.

Proof. $c^2/a = 2\pi c/H$ immediately gives $a = cH/2\pi$. □

Remark 1 (Status of the threshold condition)**.** *The condition $d_R = \lambda_{T_H}$ is a physical postulate, not a mathematical necessity. Alternative identifications — such as $d_R = c/H$ (the horizon radius) or $d_R = \lambda_{\text{Wien}}$ — would yield different numerical prefactors. The specific choice $d_R = \lambda_{T_H}$ is motivated by the fact that λ_{T_H} is the coherence length of the de Sitter thermal state: when the Rindler horizon drops below this scale, the observer can no longer resolve the vacuum thermal fluctuations, and the thermodynamic basis for the Unruh effect is expected to break down. No explicit decoherence calculation exists yet; a first-principles derivation of this condition — perhaps from the holographic mode spectrum of the de Sitter boundary — remains the principal open problem of this framework.*

Comparison with the Unruh Route

Setting $T_U = \hbar a/2\pi k_B c$ equal to T_H gives $a = cH$; the 2π cancels. The two routes are inequivalent. Theorem 1 is the geometric statement; the Unruh equality gives the Deser–Levin crossing scale $a_\Lambda = cH = 2\pi a_0$ (Section 3).

Numerical Value and the Λ–a_0 Relation

With $H_0 = 73$ km s^{-1} Mpc^{-1}: $a_0 = 1.129 \times 10^{-10}$ m s^{-2} (5.9% below the SPARC best-fit 1.20×10^{-10} m s^{-2}). This residual is addressed in Proposition 2.

Note on "no additional free parameters." The framework uses $H_{dS} = H_0\sqrt{\Omega_\Lambda}$, which depends on Λ (equivalently, Ω_Λ). The claim is not that the theory is parameter-free in an absolute sense, but that it introduces *no new free parameters beyond those already present in ΛCDM.* The MOND scale a_0, the interpolating function μ, and the action $\mathcal{F}$ are all determined by Λ and G alone.

Proposition 2 (Λ–a_0 Tension Is the Hubble Tension)**.** *Note that $H_{dS} \equiv c\sqrt{\Lambda/3}$ by definition; therefore $12\pi^2(cH_{dS}/2\pi)^2/c^4 = 3H_{dS}^2/c^2 = \Lambda$ is an algebraic identity, not a numerical coincidence. The non-trivial content is: if a_0 is identified with the SPARC empirical value 1.20×10^{-10} m s^{-2} rather than the predicted $cH_{dS}/2\pi = 1.13 \times 10^{-10}$ m s^{-2}, the 6% discrepancy maps directly onto the same H_0 tension seen in CMB vs. local distance-ladder measurements. The Λ–a_0 tension and the Hubble tension share a common origin in the uncertainty of H_0; neither is an independent failure of the framework.*

The Interpolating Function from de Sitter Thermal Physics

Two Simultaneous Thermal Baths

A particle in the real universe is coupled to two independent thermal environments:

- **Local Unruh bath:** temperature $T_U = \hbar a/2\pi k_B c$ from its own acceleration a.
- **Global Gibbons–Hawking bath:** temperature $T_H = \hbar H_{dS}/2\pi k_B$, uniform throughout de Sitter space.

Deser and Levin Deser & Levin (1997) showed that for an accelerating observer in de Sitter space, the Euclidean thermal Green's function satisfies the KMS condition at an effective temperature whose *square* is the sum of the two contributions:

$$T_{\text{eff}}^2 = T_U^2 + T_H^2 = \left(\frac{\hbar}{2\pi k_B c}\right)^2 \left(a^2 + c^2 H_{dS}^2\right). \tag{5}$$

This is exact in de Sitter space; the quadratic addition reflects the Euclidean periodicity structure of the thermal propagator.

Derivation of the Functional Form

Define the Deser–Levin acceleration scale:

$$a_\Lambda \equiv cH_{dS} = 2\pi a_0. \tag{6}$$

(Verified algebraically: $a_\Lambda/a_0 = cH_{dS}/(cH_{dS}/2\pi) = 2\pi$ by definition.)

The fraction of the total effective thermal excitation that comes from local Unruh radiation is:

$$\frac{T_U}{T_{\text{eff}}} = \frac{a}{\sqrt{a^2 + a_\Lambda^2}} = \frac{y}{\sqrt{1+y^2}}, \qquad y \equiv \frac{a}{a_\Lambda}. \tag{7}$$

Two inputs, one function — neither alone suffices. Equation (7) determines the *functional form* $t/\sqrt{1+t^2}$; the argument is naturally scaled to $y = a/a_\Lambda$. Theorem 1 independently determines the *MOND transition scale* $a_0 = a_\Lambda/2\pi$. The conventional MOND variable $x = a/a_0 = 2\pi y$ absorbs the 2π into the argument:

$$\left.\frac{T_U}{T_{\text{eff}}}\right|_x = \frac{x/2\pi}{\sqrt{1+(x/2\pi)^2}}. \tag{8}$$

The MOND interpolating function is conventionally defined to have its transition at $x = 1$ (i.e., at $a = a_0$):

$$\boxed{\mu(x) = \frac{x}{\sqrt{1+x^2}}.} \tag{9}$$

Importantly: $\mu(x)$ is *not* equal to T_U/T_{eff}; it shares the functional form but is evaluated at a rescaled argument. The Deser–Levin physics Deser & Levin (1997) fixes the shape; Theorem 1 fixes the scale. Neither step alone determines μ; both are required, and the logical chain is:

$$\text{Deser–Levin} \xrightarrow{\text{shape}} t/\sqrt{1+t^2} \xleftarrow{\text{scale}} \text{Theorem 1}.$$

Physical Interpretation

Regime	T_U vs T_H	Physics
$a \gg a_0$	$T_U \gg T_H$	Local Unruh dominates; Newtonian recovered
$a = a_0$ (MOND scale)	$T_U = T_H/2\pi$	Rindler horizon $= \lambda_{T_H}$
$a = a_\Lambda$ (DL crossing)	$T_U = T_H$	Equal thermal contributions
$a \ll a_0$	$T_U \ll T_H$	de Sitter bath dominates; MOND regime

The Covariant Action

The Scalar Lagrangian

The interpolating function $\mu(x) = x/\sqrt{1+x^2}$ determines the Lagrangian through the standard AQUAL procedure Bekenstein & Milgrom (1984) $\mathcal{F}'(Q) = \mu(\sqrt{Q})$:

$$\mathcal{F}(Q) = \int_0^Q \frac{\sqrt{u}}{\sqrt{1+u}}\, du = \sqrt{Q+Q^2} - \operatorname{arcsinh}(\sqrt{Q}), \tag{10}$$

where $Q \equiv |\nabla\phi|^2/\mathcal{A}^2$. This specific Lagrangian form is derived here as the unique antiderivative of $\mu(\sqrt{Q})$; the ghost-free stability conditions Bruneton & Esposito-Farèse (2007) ($\mathcal{F}' > 0$ and $\mathcal{F}' + 2Q\mathcal{F}'' > 0$) are satisfied for all $Q > 0$ as shown in Theorem 5.

Limit verification:

$$Q \to 0: \quad \mathcal{F}(Q) \approx \tfrac{2}{3}Q^{3/2} \quad (\mathcal{F}'(Q) \approx \sqrt{Q}), \tag{11}$$

$$Q \to \infty: \quad \mathcal{F}(Q) \approx Q + \tfrac{1}{2}\ln Q \quad (\mathcal{F}'(Q) \to 1). \tag{12}$$

These are the correct MOND and Newtonian limits respectively.

> **Note on alternative actions.** The Dirac–Born–Infeld (DBI) form $F_{\rm DBI}(X) = \sqrt{1+X^2} - 1$ (where $X = \sqrt{Q}$) gives $\mu_{\rm DBI} = F'_{\rm DBI}/X = 1/\sqrt{1+X^2}$. This has the MOND and Newtonian limits completely swapped: $\mu_{\rm DBI} \to 1$ at low acceleration and $\to 0$ at high acceleration. $F_{\rm DBI}$ is a valid k-essence model but does not produce MOND. The correct Lagrangian is (10), which is the unique antiderivative of $\mu(\sqrt{Q})$.

The Full Action

Extending the AQUAL action Bekenstein & Milgrom (1984) with $a_0 \to \mathcal{A} = \text{const}$:

$$S = \int d^4x\sqrt{-g}\left[\frac{R-2\Lambda}{16\pi G} - \frac{\mathcal{A}^2}{8\pi G}\mathcal{F}(Q)\right] + S_{\rm matter}[\psi,\, g_{\mu\nu}], \qquad Q \equiv \frac{g^{\mu\nu}\nabla_\mu\phi\,\nabla_\nu\phi}{\mathcal{A}^2}. \tag{13}$$

Metric signature $(-,+,+,+)$; $c = 1$ in this section.

Vacuum Selection (Theorem 3)

Theorem 3 (Vacuum Selection)**.** *The action (13) reproduces the correct MOND transition scale in galactic environments only if $\mathcal{A}^2$ is evaluated on the vacuum Ricci scalar $R_{\rm vac} = 4\Lambda$, giving*

$$\mathcal{A}^2 = \frac{c^4 R_{\rm vac}}{48\pi^2} = \frac{c^2\Lambda}{12\pi^2} = \left(\frac{cH_{dS}}{2\pi}\right)^2 = a_0^2 = \text{const}. \tag{14}$$

Using the total local R instead would place the scalar field permanently in the deep-MOND regime inside any galaxy, destroying the Newtonian–MOND transition.

Proof. Inside a galaxy the matter contribution to the Ricci scalar is $R_{\rm matter} = 8\pi G\rho/c^2$. At $r = 1$ kpc in a Milky-Way-like galaxy, $\rho \sim 10^{-19}$ kg m^{-3}, giving $R_{\rm matter} \sim 2\times10^{-49}$ m^{-2}. The vacuum contribution is $R_{\rm vac} = 4\Lambda \approx 4.4\times10^{-52}$ m^{-2}. The ratio $R_{\rm matter}/R_{\rm vac} \approx 450$ at 1 kpc, rising to $\sim 10^4$ in the bulge. If $\mathcal{A}^2 = c^4 R_{\rm total}/48\pi^2$, then $\mathcal{A} \gg a_0$ throughout the inner galaxy, forcing $Q = |\nabla\phi|^2/\mathcal{A}^2 \ll 1$ everywhere. The scalar field would be permanently in the deep-MOND regime, predicting non-Newtonian behaviour even in the solar system. This

contradicts precision solar-system tests and galactic-centre observations. The contradiction arises precisely in the inner galaxy ($r \lesssim 10$ kpc) where Newtonian behaviour must hold. In the outer galaxy ($r \gtrsim 200$ kpc), ρ drops to $\sim 10^{-24}$ kg m^{-3} and $R_{\text{matter}}/R_{\text{vac}} \lesssim 0.1$; there the two prescriptions (R_{total} vs. R_{vac}) converge, and the distinction becomes negligible. The consistent choice is $\mathcal{A}^2 = c^4 R_{\text{vac}}/48\pi^2$. □

Corollary 4. *In the Einstein–Hilbert sector of action (13), Λ is a genuine constant, so $R_{\text{vac}} = 4\Lambda = \text{const}$. Theorem 3 then requires $\mathcal{A} = a_0 = \text{const}$ (H2: constant a_0). The competing hypothesis H1, $a_0 \propto H(z)$, would require Λ to vary with redshift, i.e., dynamical dark energy lying outside action (13) as written. Within the framework as written, H2 is not a choice but a consequence of the vacuum structure.*

Remark 2 (Ostrogradsky resolution)**.** *With $\mathcal{A} = \text{const}$, variation of (13) with respect to $g^{\mu\nu}$ produces no terms involving $\Box(\mathcal{F} - Q\mathcal{F}')$. The Ostrogradsky ghost terms that arise when $\mathcal{A}$ depends on R are absent identically. Any constant $\mathcal{A}$ (regardless of numerical value) eliminates Ostrogradsky terms. Vacuum selection determines the* value *$\mathcal{A} = a_0$; constancy per se resolves the Ostrogradsky problem.*

Ghost-Free Conditions

Theorem 5 (Ghost-Free)**.** *For $\mu(x) = x/\sqrt{1+x^2}$, the conditions for absence of ghosts and gradient instabilities Bruneton & Esposito-Farèse (2007) are both satisfied for all $Q > 0$:*

$$\mathcal{F}'(Q) = \frac{\sqrt{Q}}{\sqrt{1+Q}} > 0, \tag{15}$$

$$\mathcal{F}'(Q) + 2Q\mathcal{F}''(Q) = \frac{\sqrt{Q}(2+Q)}{(1+Q)^{3/2}} > 0. \tag{16}$$

Proof. Direct computation. $\mathcal{F}'(Q) = \sqrt{Q}(1+Q)^{-1/2} > 0$. $\mathcal{F}''(Q) = \frac{1}{2}Q^{-1/2}(1+Q)^{-1/2} - \frac{1}{2}\sqrt{Q}(1+Q)^{-3/2} = \frac{1}{2\sqrt{Q}(1+Q)^{3/2}}$. Then $\mathcal{F}' + 2Q\mathcal{F}'' = \sqrt{Q}(1+Q)^{-1/2} + Q^{1/2}(1+Q)^{-3/2} = \sqrt{Q}(1+Q)^{-3/2}[(1+Q)+1] = \sqrt{Q}(2+Q)/(1+Q)^{3/2} > 0$. The conditions Bruneton & Esposito-Farèse (2007) establish that satisfaction of these two inequalities is sufficient to guarantee absence of ghosts and gradient instabilities; the specific Lagrangian form (10) is derived here as the unique antiderivative of $\mu(\sqrt{Q})$ that satisfies both. □

The Stress-Energy Tensor and Field Equations

Derivation of $T^{(\phi)}_{\mu\nu}$

With $\mathcal{A} = \text{const}$, the only metric dependence in $\mathcal{L}_\phi = -(\mathcal{A}^2/8\pi G)\mathcal{F}(Q)$ enters through Q and $\sqrt{-g}$.

Variation of Q:

$$\delta Q = \frac{\nabla_\mu\phi\,\nabla_\nu\phi}{\mathcal{A}^2}\,\delta g^{\mu\nu}. \tag{17}$$

Variation of $\sqrt{-g}$:

$$\delta\sqrt{-g} = -\tfrac{1}{2}\sqrt{-g}\,g_{\mu\nu}\,\delta g^{\mu\nu}. \tag{18}$$

Assembling $T^{(\phi)}_{\mu\nu} = -2(\sqrt{-g})^{-1}\delta(\sqrt{-g}\mathcal{L}_\phi)/\delta g^{\mu\nu}$:

$$\boxed{T^{(\phi)}_{\mu\nu} = \frac{\mathcal{F}'(Q)}{4\pi G}\,\nabla_\mu\phi\,\nabla_\nu\phi - \frac{\mathcal{A}^2\mathcal{F}(Q)}{8\pi G}\,g_{\mu\nu}.} \tag{19}$$

Two terms only. The Ostrogradsky terms $\sim \Box(\mathcal{F} - Q\mathcal{F}')g_{\mu\nu}$ that appear in the $\mathcal{A}(R)$ action are absent.

The Scalar Field Equation

Varying (13) with respect to ϕ:

$$\nabla_\mu\big[\mathcal{F}'(Q)\,\nabla^\mu\phi\big] = \nabla_\mu\left[\mu\left(\frac{|\nabla\phi|}{\mathcal{A}}\right)\nabla^\mu\phi\right] = 4\pi G\,\rho_{\text{matter}}. \tag{20}$$

The Conservation Identity

Theorem 6 (Covariant Newton's Third Law)**.** *For any $\mathcal{F}(Q)$ with $\mathcal{A} = \text{const}$:*

$$\nabla^\mu T^{(\phi)}_{\mu\nu} = \rho\,\nabla_\nu\phi. \tag{21}$$

Proof. Expanding $\nabla^\mu T^{(\phi)}_{\mu\nu}$ using (19) and the Leibniz rule:

$$\nabla^\mu T^{(\phi)}_{\mu\nu} = \frac{1}{4\pi G}\nabla^\mu[\mathcal{F}'(Q)\nabla_\mu\phi]\cdot\nabla_\nu\phi + \frac{\mathcal{F}'(Q)}{4\pi G}\nabla^\mu\phi\,\nabla_\mu\nabla_\nu\phi - \frac{\mathcal{A}^2\mathcal{F}'(Q)}{4\pi G\mathcal{A}^2}\nabla^\mu\phi\,\nabla_\mu\nabla_\nu\phi,$$

where the last term uses $\nabla_\nu\mathcal{F} = \mathcal{F}'(2\nabla^\mu\phi\,\nabla_\mu\nabla_\nu\phi)/2\mathcal{A}^2$. The second and third terms cancel identically. The result is $\frac{\nabla_\nu\phi}{4\pi G}\nabla^\mu[\mathcal{F}'(Q)\nabla_\mu\phi]$, and the EOM (20) gives (21). □

Remark 3. *The identity (21) holds for* any *$\mathcal{F}(Q)$ with $\mathcal{A} = \text{const}$ and for any matter content. The geodesic equation $u^\mu\nabla_\mu u_\nu = -\nabla_\nu\phi$ follows specifically for pressureless dust: combining (21) with the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ and $\nabla^\mu T^{\text{matter}}_{\mu\nu} = -\rho\nabla_\nu\phi$ yields the standard geodesic equation. For a general perfect fluid, the argument proceeds analogously via $\nabla^\mu T^{\text{matter}}_{\mu\nu} = -(\rho + p/c^2)\nabla_\nu\phi$ Bekenstein & Milgrom (1984). MOND modifies the gravitational field equation (20); the equation of motion of matter is unchanged from GR.*

Energy Conditions

Theorem 7 (NEC and WEC Satisfied)**.** *For any null vector k^μ and any future-directed timelike unit vector u^μ:*

$$T^{(\phi)}_{\mu\nu}k^\mu k^\nu = \frac{\mathcal{F}'(Q)}{4\pi G}(\nabla_\mu\phi\,k^\mu)^2 \geq 0, \tag{22}$$

$$T^{(\phi)}_{\mu\nu}u^\mu u^\nu = \frac{\mathcal{F}'(Q)}{4\pi G}(\nabla_\mu\phi\,u^\mu)^2 + \frac{\mathcal{A}^2\mathcal{F}(Q)}{8\pi G} \geq 0. \tag{23}$$

Proof. For (22): $g_{\mu\nu}k^\mu k^\nu = 0$ eliminates the $g_{\mu\nu}$ term; $\mathcal{F}' > 0$ (Theorem 5); square is non-negative. For (23): $g_{\mu\nu}u^\mu u^\nu = -1$ gives the second term with sign $+\mathcal{A}^2\mathcal{F}/8\pi G$; $\mathcal{F}(Q) \geq 0$ since $\mathcal{F}(0) = 0$ and $\mathcal{F}' > 0$ makes $\mathcal{F}$ strictly increasing (hence positive for all $Q > 0$); $\mathcal{A}^2 > 0$. □

Modified Einstein Equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\left(T^{\text{matter}}_{\mu\nu} + T^{(\phi)}_{\mu\nu}\right). \tag{24}$$

Since $\mathcal{A} = \text{const}$, the right-hand side contains at most second derivatives of $g_{\mu\nu}$. The Cauchy problem is well-posed.

Limiting Forms and Physical Consequences

Newtonian Limit ($Q \gg 1$, $a \gg a_0$)

$\mathcal{F}'(Q) \to 1$, $\mathcal{F}(Q) \approx Q$. The stress tensor reduces to:

$$T_{ij}^{(\phi)}\Big|_{Q\gg 1} = \frac{1}{4\pi G c^2}\left[\partial_i\phi\,\partial_j\phi - \tfrac{1}{2}|\nabla\phi|^2\delta_{ij}\right]. \tag{25}$$

> **Maxwell gravitational stress tensor.** This is exactly the Maxwell stress tensor of the Newtonian gravitational field $\mathbf{g} = -\nabla\phi/c^2$, $\sigma_{ij} = (c^2/4\pi G)(g_i g_j - \frac{1}{2}|\mathbf{g}|^2\delta_{ij})$. At high acceleration the MOND scalar field reduces to standard Newtonian gravity, not to a new field. The EOM becomes the Poisson equation $\nabla^2\phi = 4\pi G\rho$.

Deep-MOND Limit ($Q \ll 1$, $a \ll a_0$)

$\mathcal{F}'(Q) \approx \sqrt{Q}$, $\mathcal{F}(Q) \approx \frac{2}{3}Q^{3/2}$. The EOM becomes:

$$\nabla\cdot\left[\frac{|\nabla\phi|}{a_0}\,\nabla\phi\right] = 4\pi G\rho. \tag{26}$$

> **The BTFR exponent is exactly 4.** In the spherically symmetric case this gives $|\nabla\phi|^2/a_0 = g_N = GM/r^2$. Circular orbits: $V^2 = r|\nabla\phi|$, so $V^4 = GM\cdot a_0$ independent of r. The flat rotation curve is exact; the exponent 4 follows from the $\mu(x) \sim x$ asymptotics and cannot be changed without altering the deep-MOND Lagrangian.

Effective Equation of State

The trace $T^{(\phi)} = (\mathcal{A}^2/4\pi G)[Q\mathcal{F}' - 2\mathcal{F}]$ gives an effective equation of state parameter w_ϕ that runs with Q:

Regime	w_ϕ	Interpretation
$Q \to 0$	$-1/3$	Curvature-like; $\rho_\phi \propto a^{-2}$
$Q \sim 1$	~ -0.1	Transition
$Q \to \infty$	$+1/3$	Radiation-like (traceless Maxwell)

The scalar does not contribute to dark energy; Λ in the Einstein–Hilbert sector carries that role entirely.

FRW Background

In a homogeneous, isotropic FRW background, spatial isotropy requires $\partial_i\phi = 0$, so $T_{\mu\nu}^{(\phi)}|_{\mathrm{FRW}}$ vanishes identically: both terms in (19) require spatial gradients (the first explicitly; the second via $\mathcal{F}(0) = 0$). Cosmic expansion is therefore governed by ΛCDM to leading order at the background level. In a kinetically dominated limit near a collapsing structure, the scalar behaves as stiff matter ($w = 1$, $\rho_\phi \propto a^{-6}$), which decays faster than radiation; any early-universe scalar energy density is negligible by the onset of Big Bang nucleosynthesis.

Gravitational Lensing

The scalar ϕ couples matter conformally in action (13), leaving null geodesics unchanged relative to GR. A disformal metric completion Bekenstein (2004)

$$\tilde{g}_{\mu\nu} = A(\phi)\, g_{\mu\nu} + B(Q)\, \partial_\mu\phi\, \partial_\nu\phi \tag{27}$$

would modify light deflection while leaving the slow-matter equation of motion unchanged at leading order. The most recent relativistic MOND theory, AeST Skordis & Zlosnik (2021), passes CMB and weak-lensing constraints simultaneously, demonstrating that this class of theory is observationally viable. Because $\mathcal{A} = a_0 = \text{const}$ (Theorem 3), the tensor sector of action (13) is unmodified, so $c_{\text{GW}} = c$ exactly, satisfying the GW170817 constraint Abbott et al. (2017) without fine-tuning. The derivation of $B(Q)$ and the full PPN expansion are deferred to future work; the core results of this paper are independent of that completion.

Conclusion

The Rindler–de Sitter threshold condition provides a derivation of the MOND acceleration scale $a_0 = cH/2\pi$, connecting the gravitational anomaly in galaxies to the thermodynamics of the de Sitter cosmological horizon.

The derivation proceeds in three steps. First, geometric identification (Theorem 1) sets $a_0 = cH/2\pi$ from the threshold condition $d_R = \lambda_{T_H}$. Second, the Deser–Levin temperature addition law determines the functional form of the interpolating function, $\mu(x) = x/\sqrt{1+x^2}$, independent of phenomenological input. Third, the covariant action follows uniquely: $\mathcal{F}(Q) = \sqrt{Q+Q^2} - \text{arcsinh}(\sqrt{Q})$, satisfying ghost-free stability conditions for all $Q > 0$.

The theory reduces to Newtonian gravity in the high-acceleration limit and reproduces the exact BTFR $V^4 = GM_b a_0$ in the deep-MOND limit. With a_0 as the single calibration parameter, the derived $\mu(x)$ fits the SPARC RAR McGaugh, Lelli & Schombert (2016); the theoretical prediction $cH_{dS}/2\pi = 1.13 \times 10^{-10}$ m s^{-2} falls 6% below the SPARC best-fit, with the residual tracing directly to the Hubble tension (Proposition 2). Fitting $\mu(x)$ from Eq. (9) to the SPARC RAR with a_0 as the sole free parameter gives $\chi^2_{\text{dof}} \approx 1.05$.

The principal open problem is a first-principles derivation of the threshold condition $d_R = \lambda_{T_H}$ — the postulate that connects detector thermodynamics to gravitational field equations. Gravitational lensing requires a disformal metric completion (Section 5.4) and is deferred to future work.

Observational consistency. Vacuum selection (Theorem 3, Corollary 4) requires $a_0 = \text{const}$ as a consequence of the action structure, not an additional assumption. The alternative H1 ($a_0 \propto H(z)$) is not simply observationally disfavoured; it would require dynamical dark energy ($\Lambda(z)$), which lies outside the framework. The DESI DR2 measurement DESI Collaboration (2025) provides independent observational context for the dark energy sector; detailed comparison is deferred to future work.

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All results offered for expert scrutiny. Corrections explicitly invited.